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Delivery tray for food product printer

Abstract

A delivery tray, system, and method for automatically delivering food product to a printer for printing on a surface of the food product. The tray has a main body having a plurality of food product support surfaces spaced apart on the main body and each food product support surface configured for supporting one of a plurality of food products thereon and the main body further comprising at least one positioning mechanism corresponding to each of the plurality of food product support surfaces and each positioning mechanism comprising at least one positioning element for guiding the loading of at least one of the plurality of food products onto the delivery tray. The tray is coupled to a motorized base for automatic movement of the tray.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present application claims priority to and the benefit of U.S. provisional application Ser. No. 62/894,323, filed on Aug. 30, 2019, the content of which is hereby incorporated in its entirety.

BACKGROUND

(1) Printing with edible ink on food products is done for purposes of identifying, marking, decorating, or personalizing the food product. For example, food products such as cookies can be passed under a print head ejecting edible ink on the top surface of the cookie, whether iced or plain, and printed with a selected design.

(2) Currently, a conveyor belt system or tray is used where one or more cookies are spaced apart in an array on the conveyor belt or tray and passed below the print head of the printer. Alternatively, labels comprised of edible paper can be printed with edible ink and applied to a food item with an

edible adhesive such as frosting or icing.

(3) Systems of the prior art that are able to print images on an array of food products supported on a conveyor belt or a tray with two or more columns of food products requires a larger and more expensive printer on the scale of a commercial printer. Such systems also usually require sophisticated sensing systems to locate the position of each food item for printing.

(4) In the prior art systems, the loading and unloading of a food product on a tray or conveyor during printing is generally impractical. The conveyor or tray are often not accessible while products are being printed. It is also difficult to accurately load a food product on a moving conveyor or tray. It is also difficult to load a food product on a moving conveyor or tray without adversely affecting the print quality of the image currently being printed.

SUMMARY

(5) An aspect of the present disclosure relates to a delivery tray for automatically delivering food product to a printer for printing on a surface of the food product. The tray has a main body having a plurality of food product support surfaces spaced apart on the main body and each food product support surface is configured for supporting one of a plurality of food products thereon. The main body further has at least one positioning mechanism corresponding to each of the plurality of food product support surfaces and each positioning mechanism comprising at least one positioning element for guiding the loading of at least one of the plurality of food products onto the delivery tray.

(6) In one or more embodiments, the at least one positioning element of each positioning mechanism has a $\frac{1}{8}$ inch radius at a joint between the positioning element and the main body of the delivery tray and wherein the at least one positioning element protrudes upwardly from a top surface of the main body.

(7) In one or more embodiments, each positioning mechanism comprises two positioning elements provided in a pair.

(8) In one or more embodiments, the food product support surfaces comprise one or more support sides for holding one or more sides of the food product and an opening therebetween, where at least one positioning element is provided on at least one of the one or more support sides for each support surface.

(9) The delivery tray is constructed from a food-grade material comprising a metal such as stainless steel, a plastic material, or a combination thereof.

(10) The delivery tray also has one or more mechanisms for engaging with a drive mechanism of a tray base for moving the delivery tray between adjacent food product support surfaces.

(11) In one or more embodiments, the one or more mechanisms for engaging with the drive mechanism of the tray base comprise one or more receiving configured to removably couple to one or more drive posts of the drive mechanism of the elements tray base.

(12) In one or more embodiments, a center locating element on the tray is provided and configured to removably couple to a locating element of the tray base for laterally locating the tray connected to the tray base.

(13) In one or more embodiments, the one or more mechanisms for engaging with a drive mechanism of a tray base comprise one or more concave surfaces, apertures, or combinations thereof for removably coupling to the one or more drive posts of the drive mechanism of the tray base.

(14) Another aspect of the present disclosure relates to a system for printing on a surface of a plurality of food products. The system has a printer configured to print on a surface of a food product and a delivery tray spaced apart from the printer and having a plurality of food product holders spaced apart on the delivery tray and each holder configured for supporting one of the plurality of food products thereon and a positioning mechanism for loading food products onto the tray and positioning the food product accurately on each holder. The delivery tray is configured for movement such that the delivery tray automatically and continuously delivers unprinted food

product to the printer for printing on a surface thereof.

(15) A base is provided for operably supporting the delivery tray in a position spaced apart from the printer.

(16) In one or more embodiments, the delivery tray is a rotatable delivery tray.

(17) Each holder of the delivery tray has one or more support sides for holding one or more sides of the food product and an opening therebetween exposing a bottom surface of the food product.

(18) The plurality of food products are edible elements having a substantially flat surface, cookies, frosting sheets, rice paper, or combinations thereof.

(19) Movement of the delivery tray about the tray base is synchronized with the printing of the food products in a sequential manner.

(20) Yet another aspect of the present disclosure relates to positioning a first food product on a first food product support surface of a delivery tray and guiding the first food product onto a selected position of the first support surface, the selected position bounded on one or more sides by a positioning mechanism located adjacent the first food product support surface of the delivery tray. The delivery tray is automatically advanced from the first food product support surface to a second food product support surface and a second food product is positioned on the second food product support surface of the delivery tray along with guiding the second food product onto a selected position of the second support surface, the selected position bounded on one or more sides by a positioning mechanism located adjacent the second food product support surface of the delivery tray.

(21) In one or more embodiments, the delivery tray is coupled to a motorized base for automatically advancing the delivery tray wherein one or more receiving elements on the delivery tray couple to corresponding elements on the motorized base for locating and leveling the delivery tray on the motorized base.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a food product printing system.

(2) FIG. 2 is a perspective view of a food product printing system with a cover portion opened for exposing interior portions of a printer portion of the system.

(3) FIG. 3 is a top perspective view of the food product printing system having an automatic delivery tray component removed and a cover of a printer removed.

(4) FIG. 4 is a side view of the food product printing system with the automatic delivery tray component removably secured to a tray base portion.

(5) FIG. 5 is a perspective view of the tray base portion.

(6) FIG. 6 is a top view of the tray secured on the base.

(7) FIG. 7 is a bottom view of the tray secured on the base.

(8) FIG. 8 is a top view of the food product printing system with food product for printing loaded thereon.

DETAILED DESCRIPTION

(9) A food product printing system of the present disclosure includes a printer system for printing an image on a surface of the food product. The print surface may be an actual surface of the food product itself. The printer system comprises a printer having a receiving arm and a delivery tray supported on a base. The system may also be provided with a plurality of different delivery trays, each tray configured for continuous delivery of a varied number of food products, and/or for food products of different or specific dimensions.

(10) The receiving arm is configured for receiving a food product from the delivery tray and moving the food product from the delivery tray into the printer and into a printing position. The

printing position is substantially below a print head of the printer such that the surface to be printed on is in the path of the print head. Once the food product is printed, the receiving arm then returns the printed food product to the delivery tray. The delivery tray is also configured to move in one or more directions to continuously and automatically deliver a plurality of food items sequentially to the receiving arm.

(11) One embodiment of the printing system **10** is illustrated generally in FIGS. **1-8**. The printing system **10** is configured to receive a food product **12** from a delivery tray **22**, to print on a surface of the food product **12** and return the food product **12** to the delivery tray **22**. The system is also configured to do so automatically and continuously for a pre-determined number of food items. The system **10** comprises a printer **14** having a print head **16** and nozzles (not shown) wherein the print head **16** and nozzles are configured for printing with edible ink. The printer **14** also supports an extendable and retractable receiving arm **20**.

(12) The receiving arm **20** is operably connected to a power source within the printer **14**. The receiving arm **20** is extendable and retractable with respect to the position of the path of the print head **16**. The receiving arm **20** has a first end and a second opposing end and a substantially flat receiving surface extending therebetween. The first end is operably connected to a mechanism for retracting and extending the arm **20**. The second end and at least a portion of the receiving surface of the receiving arm **20** are configured for extending outwardly from the printer **14**. When extended, the receiving arm **20** may receive a food product, hold a food product, and/or return a printed food product to the delivery tray **22**. The receiving arm also holds or supports the food product within the printer **14** and during printing on the surface of the food product **12**.

(13) With respect to the printer **14**, the print head **16** moves linearly back and forth along a gantry in the manner of a standard inkjet printer for ejecting (edible) ink towards the print surface and thus printing pre-selected content on a print surface positioned below the path of the print head **16**. The printer **14** is in communication with a controller wherein software may be used to control printing of the content on one or a plurality of food products.

(14) In further detail, the receiving arm **20** moves in a direction perpendicular to the print head path such that the receiving arm **20** extends and retracts across or through the path of the print head **16**. In one embodiment, the receiving arm **20** is a substantially flat, rectangular support plate which extends towards a holder **26** of the delivery tray **22** in order to engage with and remove the food product **12** from the delivery tray **22** as well as to return printed food products to the delivery tray. A top surface of the receiving arm **20** is a receiving surface for food product and may be textured or otherwise provided with a surface to increase a co-efficient of friction between the receiving arm **20** and the food product for retaining the food product on the arm **20**.

(15) The receiving arm extends to receive or return the food product **12** and retracts from the tray **22** into the printer **14** and into a position below the print head **16** for printing. Once content is printed on a food product, the receiving arm **20** then extends to return the food product **12** back to the delivery tray **22**. The delivery tray **22** then advances to provide a second unprinted food product **12** to the receiving arm **20**. The receiving arm **20** is also configured for vertical movement such that the receiving surface of the receiving arm **20** is movable between two positions, a raised and a lowered position, where these positions allow the receiving arm **20** to retrieve and return food products **12** with respect to the delivery tray **22**, and may allow for advancing of the delivery tray **22** when the receiving arm **20** is extended. Once loaded, the receiving arm **20** is then retracted to the printing position with the food product **12** thereon. This process continues on continuously and automatically until a pre-determined amount of food products are printed. It is also contemplated that the delivery tray may also be raised or lowered in order to allow for the delivery tray **22** to advance to a subsequent holder **26** or opening **28** on the delivery tray **22** for automatically delivering, receiving and thus printing on a plurality of food items.

(16) As illustrated in FIGS. **4** and **8**, positioned adjacent or near the printing system is a tray base **40** for providing power and controlling advancement and vertical positioning of the delivery tray

22 that is removably coupled thereto. The base **40** is spaced apart from the printer **14** and may be removably connected to the printer **14** via a connecting arm **50** that extends from a housing of the base **40** to removable connection or coupling to a housing of the printer **14**. The arm **50** also positions the base **40** at a selected distance from the printer **14** and receiving arm **20** so as to accommodate the delivery tray **22** supported thereon. The base **40** may also be operably connected to the controller and to the power supply of the printer **14** so as to control movement of the tray **22** in a manner synchronized with the printer **14**.

(17) Referring to FIGS. **6-8**, the delivery tray **22** is configured with a plurality of holders **26** as described in further detail below. Each holder **26** is configured to receive and support a food product **12** and deliver said food product **12** to the receiving arm **20**. The delivery tray illustrated includes twelve (12) holders **26**, however, trays for holding fewer or more food products are contemplated and within the scope of this disclosure. The number of holders on a tray may be adjusted based on the type or size of item being printed.

(18) In further detail, in one embodiment, each holder **26** has two support sides **24** extending from a center area of the tray **22** and an opening **28** between these support sides **24**. The food product is supported on the holder by way of resting on top of the two support sides **24** of the holder. The opening **28** between the sides **24** of the holder **26** is configured to receive the receiving arm **20** therein and allows for the receiving arm **20** to take a position for receiving and returning the food product to the delivery tray **22**. That is, the receiving arm **20** can extend into this opening **28** and contact a bottom surface of the food product **12** that is supported on the sides **24** of the holder **26**. The receiving arm **20** then supports the food product **12** thereon when retracting the food product **12** into the position for printing.

(19) It is also contemplated and within the scope of this disclosure that each holder **26** of the tray **22** may comprise a single food product support surface in lieu of two support sides **24**. Such a single food product support surface may comprise a narrow support surface for each food product, for example, in the manner of one support side **24**. Thus, food products **12** are loaded onto the single food product support surface such that the food product **12** is positioned centrally on top of the single food product support surface. In such embodiments, the receiving arm **20** may then be provided with a platform having two opposing support sides and an open space therebetween. The open space provided for accepting the single food product support surface therebetween. In such an embodiment, the receiving arm **20** and platform extend to the tray **22** and the two support sides of the platform are positioned on opposing sides of the single food product support surface of the tray **22** for transferring food products between the platform and the tray **22**.

(20) In the embodiment illustrated, each holder **26** is configured to support or otherwise hold one or more sides, edges, or perimeter lengths of the food product **12** via contact of the support sides **24** with the bottom surface of the food product **12**. The holders **26** may have one or more positioning mechanisms **30** for directing a position of the food product **12** when loaded onto the delivery tray **22**. Positioning mechanisms **30** are provided, for example, on each of the support sides **24**. Utilizing the positioning mechanisms **30** to set the position of the food product **12** during loading of the delivery tray **22** ensures delivery of the food product **12** to the receiving arm **20** such that the food product is balanced on and supported by the receiving arm **20** and ensures the surface of the food product is centered for printing of the content on the food product.

(21) In one or more embodiments, the delivery tray **22** is provided with one or more positioning mechanisms **30**, which may comprise for example, one or more positioning elements **32**, which are strategically and selectively positioned on the tray **22** to allow for ease in loading and un-loading food products **12** on the delivery tray **22**. The elements **32** can be posts, arms, or other mechanisms configured to provide a visual or tactile indicator of position of a food product **12** loaded onto the tray **22**. In the non-limiting example of the illustrated embodiment, the elements **32** protrude upwardly from connection with the tray **22** and are configured to mark the placement location for a food product **12** on the delivery tray **22**. Additionally, the positioning elements **32** also serve as

guides for loading and positioning the food product **12** onto the tray, as the food product is loaded on to the tray **22**, for example. The positioning elements **32** serve as guides for fast and efficient loading and positioning of the food product when the food product is pushed, slid, or placed on the holder **26** of the delivery tray **22**. The guides limit and guide the position of the food product, preventing the food product from taking position on the tray **22** not aligned with a holder **26**. The elements **32** may be provided in pairs which protrude from a location on a corresponding one of the support sides **24** of each holder **26**. The tray **22** may comprise alternative positionings for the elements **32** in order to accommodate food products **12** of different sizes, shapes, and types. For example, the elements **32** are positioned on the tray **22** such that a forward and/or side surface of the food item contacts and abuts the elements **32** to ensure sufficient placement for delivery to the receiving arm **20** when the food product **12** is placed on the holder **26**.

(22) The positioning mechanisms **30** allow for sufficiently accurate and easy loading of the tray with unprinted food product such that food product received on the delivery tray **22** is accurately delivered to a reciprocal position on the receiving arm **20** selected for printing. In some embodiments, this can eliminate the need for a sensors or other mechanism to detect the position of the food product on the receiving arm for printing.

(23) Loading of the delivery tray **22** with unprinted food product **12** can be done manually or in an automatic manner. As such, the system can be used with a conveyor or other automatic arm or other automated delivery system for delivering unprinted food product **12** to the delivery tray **22** and/or removing the printed product **12** from the delivery tray **22**.

(24) In one or more embodiments, the positioning mechanisms **30** comprise one or more positioning elements **32** having about a $\frac{1}{8}$ " radius at the point where the element joins or meets the surface of the delivery tray which allows for easy cleaning of the tray **22** per NSF standards.

(25) The delivery tray **22** advances about the openings **28** such that the delivery tray **22** advances from one opening **28** to the next adjacent opening **28** automatically after a food product **12** has been returned to the tray **22** by the receiving arm **20**. For example, in the embodiment illustrated, the delivery tray **22** is a generally round tray that is rotatable about its center to advance the position of the holders **26** with respect to delivery of subsequent food products **12** to the receiving arm **20**. The holders **26** are spaced apart around the perimeter of the delivery tray **22**. The delivery tray **22** and base **40** may be a carousel for continuous delivery, loading and reloading of unprinted food products after printed food products have been returned to the delivery tray **22** and subsequently removed. While a rotatable tray is illustrated herein, the delivery tray **22** may also be linearly advanced, move in multiple directions, or move in the manner of a conveyor belt, by way of non-limiting examples. That is, once printed and returned to the delivery tray **22** as the delivery tray **22** advances to the next adjacent opening **28** for purposes of delivering one or more subsequent food products **12** to the printer **14**, the printed and advanced food product may be removed and a new unprinted food product **12** placed on the tray **22**. Thus, delivery of unprinted food products and receipt of said food products after printing can be essentially endless, limited only by supply or selection of a predetermined amount to be printed, for example.

(26) As described previously, the holders **26** are generally spaced apart around the perimeter of the delivery tray **22** and the holders **26** have dimensions which allow the position of the food product **12** to be loaded onto the tray **22** in a known position. When the food product **12** is delivered to the receiving arm **20**, the position of the food product **12** print surface with respect to the print head **16** is known when the receiving arm **20** is retracted. This allows for the elimination of sensors or other position or item detecting means and eliminates other delays due to location or position detection before printing. Instead, upon retraction of the receiving arm **20**, printing can begin. This allows the automated process to proceed continuously and efficiently at an increased speed.

(27) The delivery tray **22** moves about an operable connection with a base **40**. The delivery tray **22** may rotate or advance in a linear manner, or other directional movement which advances the delivery tray from one opening **28** to the next adjacent opening **28**. In one non-limiting

embodiment, as illustrated, the delivery tray **22** rotates about the operable connection with the base **40**.

(28) Referring to FIGS. 5-7, the delivery tray **22** is operably, and removably connectable to the base **40** about one or more connectors or receiving mechanisms **43** positioned on the delivery tray **22**. These connector(s) **43** may comprise one or more drive posts **47** that both drive the rotation of the delivery tray **22** and locate the delivery tray **22** relative to the base **40** in the direction of rotation. The tray **22** may additionally further comprise another receiving element **42**, which can be positioned in a center area of the tray **22**, where the receiving element **42** connects to a locating element **46** of the base **40**. In embodiments where the receiving element **42** and locating element **46** are provided, in addition to the one or more connectors **43** and drive posts **47**, the locating element **46** locates the delivery tray **22** laterally relative to the base **40**. In embodiments where the locating element **46** and corresponding receiving element **42** of the tray **22** are not present, the one or more drive posts **47** may be used to also locate the delivery tray **22** laterally relative to the base **40**.

(29) Each of the one or more drive posts **47** may be provided with a shoulder **48** that is positioned at substantially the same level or height as a plurality of support pads **49** spaced around the outside of the base **40**. For example, the support pads **49** and the shoulder **48** of each drive post **47** combine to support the delivery tray **22** in a level position on the tray base **40**. In the embodiment illustrated, four support pads **49** are provided on the base **40** and contact a surface of the tray **22** for supporting and leveling.

(30) The connectors **42** and/or **43** of the delivery tray **22** may be concave members with respect to a bottom surface of the tray and thus convex portions on a top surface of the tray **22**. These openings **42** and **43** have interior surfaces that are configured to mate with and/or receive and engage with the respective locating post **46** and/or drive posts **47**. The locating post and/or drive posts may also have tapered surfaces thereon which support proper installation of the tray **22** with ease and in a more efficient (e.g., faster) manner. Such a connection between the locating post **46** and/or drive posts **47** and the corresponding connectors **42** and/or **43** further allows the tray **22** to generally or substantially maintain rotation when inadvertently bumped.

(31) A drivetrain is provided within a housing **52** of the base **40** and is connected to the power source of the printer. The connection mechanism between the tray **22** and base **40** allows the tray to be quickly installed on the base **40** without tools. The tray **22** is configured to be placed on the base **40** with complimentary mating of the base and tray components of the connection mechanism. This system allows various delivery trays **22** of different dimensions and arrangements for holding food products **12** to moveably connect with the base **40** with the same type of connection mechanism. While the above interface is described, alternative interfaces and non-locking mechanisms for transferring movement generated at the base **40** to the tray **22** for controlled advancement of the delivery tray **22** are contemplated.

(32) The base **40** may comprise its own power source for providing rotational power that is transferred from the base **40** to the delivery tray **22**, or the base **40** may be operably connected to the power source of the printer **10**. Thus, when the delivery tray **22** is positioned on and in connection with the base **40** as described above, the tray **22** rotates as controlled by the base **40**. The base **40** is also in communication with the controller which controls extension and retraction of the receiving arm **20** and printing of an image on a received food product **12** such that the rotation of the delivery tray is synchronized with the extension and/or retraction of the receiving arm **20**.

(33) In the embodiment illustrated, the base **40** provides the power to rotate the delivery tray **22** between various positions which include alignment of a first and one or more subsequent openings **28** with the extended receiving arm **20** as well as rotating the delivery tray **22** to advance the delivery tray **22** to a next opening **28** while the receiving arm **20** is extended. In embodiments where the delivery tray **22** may be in one of two positions, the raised or lowered position, depending on whether an item is to be delivered or returned to the tray **22** and to prevent the receiving arm **20** from interfering with advancing or rotation of the delivery tray **22**, the base **40** the

also allows for moving the delivery tray 22 between the two positions and thus is configured to raise and lower the tray or tilt the tray 22 upwardly or downwardly at the opening 28 positioned for delivery or return of the food product 12. In such an embodiment, as the receiving arm 20 is extended and positioned in the opening 28, the upper surface of the receiving arm 20 may be substantially level to the top surface of the delivery tray 22. To ensure or facilitate delivery of the food product 12 supported on the delivery tray 22, the delivery tray 22 is downwardly tilted or lowered in the direction of the receiving arm 20. This allows the receiving arm 20 to contact and support a bottom surface of the food product 12 and remove support by the delivery tray 22 thereby transferring the food product 12 to the receiving arm 20. Conversely, when the receiving arm 20 is returning a food product after printing thereon, the delivery tray 22 may be provided in a lowered position and then tilted upwardly or raised after the receiving arm 20 supporting the printed-on food item is extended into opening 28. This allows the holder 26 to lift the food product off the receiving arm 20 for removal from the printing queue.

(34) In the illustrated embodiment the food product is a dessert item such as a cookie. However, while examples of food items include various shapes, sizes, and types of cookies, the food items that can be printed with this system are not so limited. Frosting sheets, rice paper, bakery goods having at least one substantially flat surface for printing an image thereon, and other food items may be printed using this system. The delivery tray may be configured with positioning mechanisms which are tailored to the specific shape, size, thickness, type, etc. of food product to be printed and thus, the system may be provided as a kit including the printer described herein including receiving arm and one or more delivery trays of various sizes, dimensions and for delivery of one or more different food products or food products having different dimensions.

(35) The system is further configured with one or more control boards in communication with a computer or controller for configuring and initiating the printing process. However, the delivery tray is configured with one or more positioning mechanisms such that in combination with the receiving arm as a food product is delivered to the printer and positioned below the print head, the need for a sensing system or series of sensors for detecting the position of the food product is eliminated. A sensor may be utilized in the printer for confirming the presence of the food product in the printer for printing. The sensors may be positioned on the print head and used to detect only the presence of the substrate for printing. The system is a fixed spot printing system so there is no need for scanning for position of the print surface. The system also allows for loading of unprinted food products and unloading of printed food products during printing or otherwise “on the fly.” The control board or boards allow for the delivery of a first food product, printing of the first food product, return of the first food product, advancing of the delivery tray to provide a second food product for delivery, delivery of the second food product, printing of the second food product, return of the second food product to the delivery tray and advancing for delivery of one or more subsequent food products for printing and return. Thus, the system allows for the automatic delivery, printing on and return of a plurality of food products.

(36) Although the present disclosure has been described with reference to preferred embodiments, workers skilled in the art will recognize that changes may be made in form and detail without departing from the spirit and scope of the disclosure.

Claims

1. A system for printing on a surface of a plurality of food products, the system comprising: a printer configured to print on a surface of a food product and having an extending arm configured to extend from and retract into a housing of the printer; and a delivery tray spaced apart from the printer and having a plurality of food product holders spaced apart on the delivery tray and each holder configured for supporting one of the plurality of food products thereon, each food product holder comprising a pair of flat surfaces spaced apart and configured to support one or more

perimeter surfaces of the food product thereon, wherein the space between each flat surface of the pair of flat surfaces is open and is configured for positioning below the food product supported on the holder and exposing a bottom surface of the food product as the space between each flat surface is configured to receive the extending arm of the printer to remove the respective food product from the delivery tray before printing thereon; and a positioning mechanism spaced apart from the pair of flat surfaces of the holder for loading food products onto the delivery tray and positioning the food product accurately on each holder, and wherein the delivery tray is configured for movement such that the delivery tray automatically and continuously delivers unprinted food product to the extending arm of the printer for printing on a surface thereof.

2. The system of claim 1, and further comprising a base for operably supporting the delivery tray in a position spaced apart from the printer.

3. The system of claim 1, wherein the delivery tray is a rotatable delivery tray.

4. The system of claim 1, wherein the delivery tray is constructed from a material comprising a metal, stainless steel, or a plastic material.

5. The system of claim 1, wherein the plurality of food products are edible elements having a flat surface, cookies, frosting sheets, rice paper, or combinations thereof.

6. The system of claim 2, wherein movement of the delivery tray about the base is synchronized with the printing of the food products in a sequential manner.
